

Hoofdstuk 2:

Lesing 2.2.

①

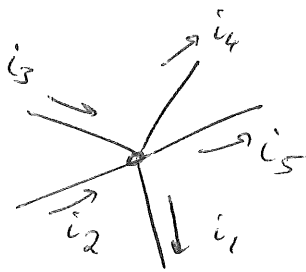
→ Kirchoff se wetten.

* KCL → Kirchoff Current Law.

→ Som van strome wat by 'n node in- en uitvloei is gelyk aan nul.

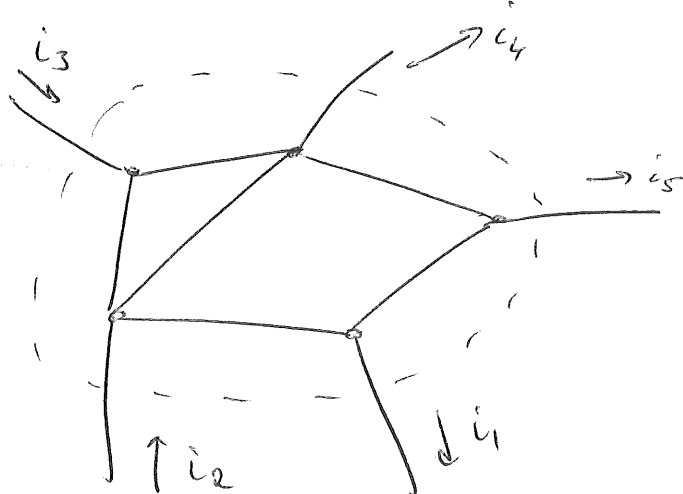
$$\sum I_{IN} = \sum I_{UIT}$$

$$\sum I_{IN} - \sum I_{UIT} = 0.$$



→ "Super"-node

Geslote grens waar stroom in en uit vloei.



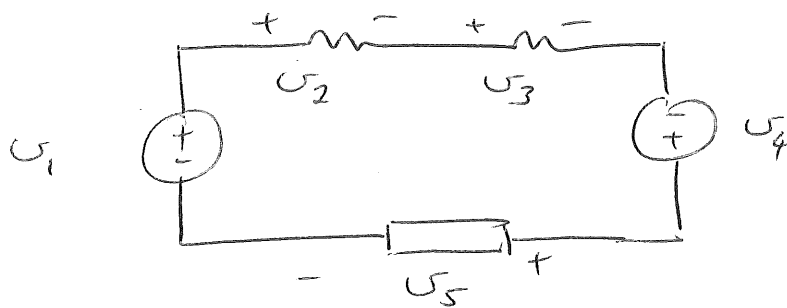
* KVL: Kirchhoff Voltage Law.

(2)

→ Som van alle spanningen in 'n geslote lus is nul

afhanklik of onafhanklik.

$$\sum V_{lus} = 0.$$



* Reëls: ① Kies rigting vir lus

② In by + beteken sommering.

③ In by - beteken aftrekking.



$$-V_1 + V_2 + V_3 - V_4 + V_5 = 0$$



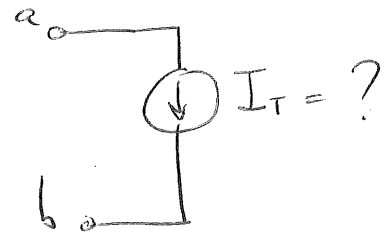
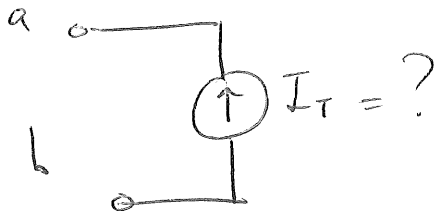
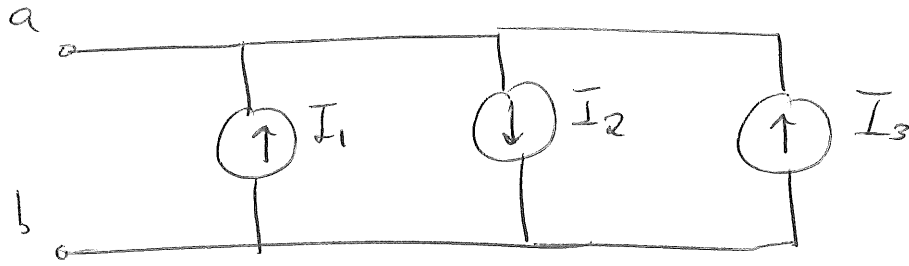
$$-V_5 + V_4 - V_3 - V_2 + V_1 = 0.$$

(Dink aan toename of afname in energie.)

→ Stroombron

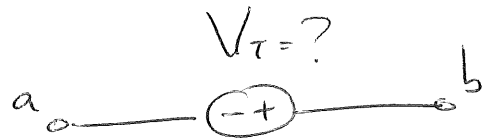
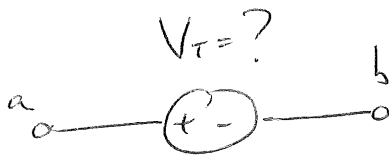
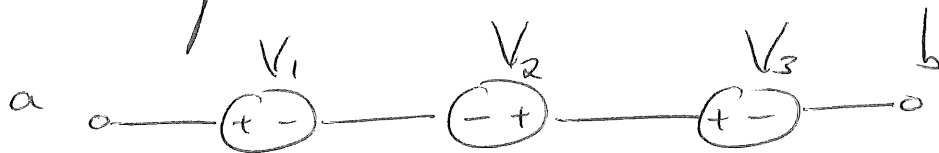
(3)

* Skryf as enkele bron.

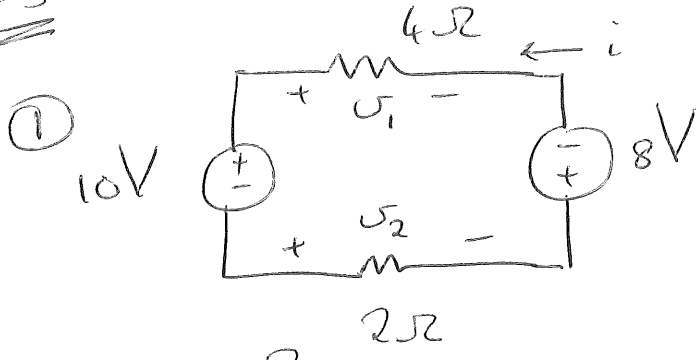


→ Spanningsbron

* Skryf as enkele bron.



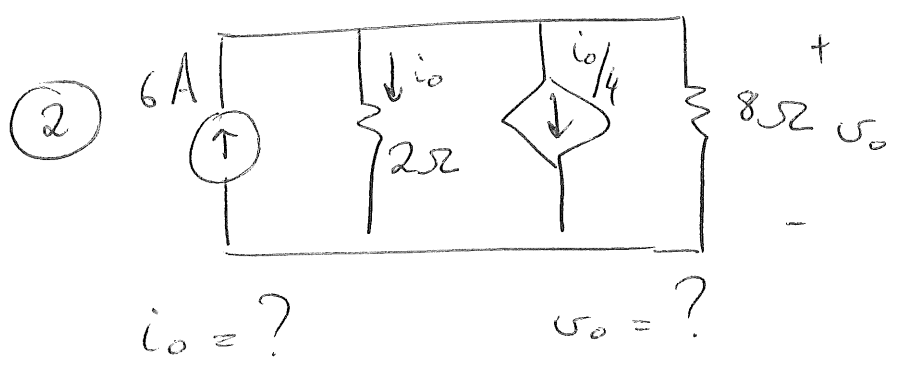
V₆



$v_1 = ?$

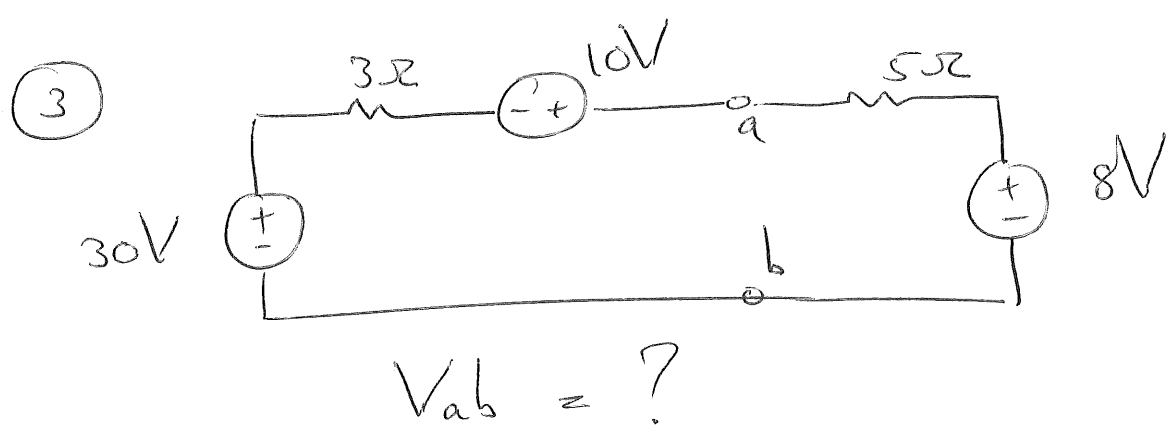
$v_2 = ?$

$i = ?$



$i_o = ?$

$v_o = ?$



$V_{ab} = ?$